

Proof Architecture in the OpenAI Lean Library: Representation, Locality, and Semantic Transport

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Abstract

Large formalizations can be mathematically revealing at the points where a proof changes representation. This source-driven review studies six such points in the OpenAI Lean mathematics collection: common-base counting, graph switch chains, the symmetric Mahler inequality, the complete Crouzeix bound, extended trace cones, and marked character-variety constructions. The cases connect finite random experiments to physical tape positions, small matrix counts to conditional expectations on graph states, ordered exterior products to differentiable fields, matrix amplification to rectangular compression, positive cuts to closed ideals, and named graded maps to a constructed surface diagram. Their common feature is an explicit preservation theorem strong enough for the next mathematical operation. We organize these correspondences by what they preserve: exact data, probability laws, quantitative estimates, or local differential information. Elementary worked examples explain why weaker correspondences can fail; the accompanying Lean examples verify the tape-index arithmetic and the recovery of finite multiplicities from observables. Comparison with established formalization and verified-programming literature locates the interest in the concrete mathematical interfaces and their composition. The resulting analysis offers a way to read substantial Lean developments as mathematical arguments and to identify interfaces useful beyond their original endpoints.

Keywords: formalized mathematics; Lean; proof architecture; semantic preservation; probabilistic algorithms; mathematical interfaces.

1 Introduction

A theorem prover checks a formal proposition, but a mathematician also wants to know why that proposition expresses the intended argument. This question becomes particularly interesting when a proof crosses several mathematical languages. A graph may become a Boolean matrix, a differential form may acquire a continuously bundled representation, or an estimator may become a program reading a particular sequence of random bits. At each crossing, the proof must identify the information that the next step consumes and show that the chosen representation retains it.

Lean provides a dependent type theory, an extensible elaborator, and a programming language [16]. Mathlib supplies a large shared mathematical vocabulary [35]. Neither a dependent record nor an automated arithmetic proof is unusual by itself. The useful question is what mathematical specification a record carries, and what argument becomes possible after that specification has been

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proved. Accounts of proof anatomy and specification-driven development make this perspective explicit [2, 12].

Substantial precedents also matter when assessing a new collection. The formalizations of the odd-order theorem and the Kepler conjecture already combined extensive libraries with large proof decompositions and verified computation [19, 20]. In Lean, perfectoid spaces and schemes required careful choices of mathematical objects and interfaces [7, 8]. Such projects show why the presence of a difficult theorem, or a large number of declarations, is insufficient evidence for a new formalization method. A useful review must locate a more specific mechanism and explain its mathematical purpose.

The OpenAI collection offers an unusually broad setting in which to look for these mechanisms [32]. We select six developments whose sources expose a substantial transition between representations. The first two concern finite probability and algorithms; the next two concern analysis; the final two concern reconstruction and transport of algebraic data. The selected statements have different roles: four are public endpoints, while trace-ideal transport and all-seams compatibility are support results with their own explicit conclusions. Reading the support results on their own terms reveals useful mathematics that a list of headline theorem titles would miss.

Three questions guide the review. What information does the intermediate representation retain? Which theorem connects it to the intended object? Where is that theorem used to discharge the next step’s hypotheses? For common bases, the answer includes the law of the output and the execution cost of one fixed machine. For Mahler, it includes both equality of form-valued fields and agreement near the integration boundary. For character varieties, it includes named maps on graded pieces, beyond equality of the corresponding subspaces.

The contribution is a comparative analysis of these concrete interfaces. We give source anchors, trace the relevant definitions and hypotheses, and separate established methods from the particular mathematical choices made in each development. We then work out small examples that expose the reason for the stronger specifications. These examples are elementary; their role is explanatory. The proposed distinction between preservation of data, laws, estimates, and local information is a reading framework, rather than a claim that the underlying methods originate in this release.

Section 2 explains the source selection and evidence. Table 1 gives the overview before the case studies in Section 3. Section 4 develops the examples, and Section 5 considers what can be reused and what the comparison establishes about novelty.

2 Sources, selection, and reading method

The source inventory is the complete Git tree of `openai/math` at commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`, inspected on 8 October 2026. It contains 122,875 files with the `.lean` extension. This is an inventory count, including supporting modules, rather than a count of independently formalized manuscripts. The separate formalization catalogue lists 173 source records and 200 main-result bindings [27]. Its metadata and the README’s manuscript totals describe different units. Our selection and conclusions concern the six developments below.

Selection was purposive. A case had to contain a mathematically meaningful change of representation, an identifiable preservation statement, and a visible use of that statement in a later argument. We favored diversity of mathematical setting and preservation requirement over coverage by file count. For each case we read backwards from the selected conclusion to its definitions and immediate hypotheses, then forwards through the concrete constructions that discharge those hypotheses. The accompanying evidence ledgers retain declaration names, line ranges, source excerpts, and Git blob identifiers. Downloaded files were checked against those identifiers.

The distinction between a local interface and its instantiation is central to this reading. A conditional analytic theorem can be a useful component even when it assumes a flux identity or a resolvent estimate. The next question is whether the endpoint constructs that input for the stated objects. Similarly, an existence witness chosen classically has a proved body, whereas a theorem with an extra provider hypothesis retains that hypothesis. We follow these distinctions rather than inferring mathematical scope from declaration names.

The collection also provides Comparator challenge configurations [29]. The separately inspected Comparator source at commit `ca04cfc72b550331658ec314bf47685281bfd4bf` implements statement comparison, dependent-declaration and axiom inspection, kernel replay, and an optional external checker [9]. This checker snapshot was pinned for our infrastructure reading; the collection’s installation instructions do not specify that revision. The architectural distinction between producing a proof and checking it has a long history [4, 14]. Work on a Lean typechecker gives a contemporary example of examining the checking implementation itself [10]. Semantic agreement between a formal statement and its mathematical interpretation remains a separate reading task.

Our evidence for the six source cases consists of static declaration and dependency analysis. Executable verification in this review concerns the original examples in Section 4, compiled with the collection’s specified Lean version, 4.34.1. This division records the scope of the evidence: the source analysis explains the selected architectures, while the small compiled examples check particular arithmetic and finite-law interfaces independently. Library maintenance and semantic linting offer complementary checks at another level [37].

Blueprints can connect informal arguments to declaration names and dependencies [25]; Lean-Architect studies automation of that connection [38]. Here the evidence ledger serves a narrower purpose: making each discussed bridge inspectable. A dependency edge alone does not explain why a hypothesis is the right one, so the exposition supplies the relevant mathematical objects and equations.

Table 1: Overview of the selected formal statements and the information preserved by each bridge.

Development	Representation bridge	Mathematical work exposed
Common bases Endpoint	Matroid transversals; finite random trees; oracle expressions; physical random tapes.	Output-law preservation joins bit-cost bounds and a fixed-program specification.
Graph switches Endpoint	Fixed-degree graph cells; 2×2 Boolean matrices; conditional averaging operators.	A finite count isolates one exceptional direction per fiber; compensated estimates control its global contribution.
Mahler Endpoint	Ordered shuffle forms; continuously bundled fields; boundary-local extensions.	Exact slot identifications preserve derivatives, signs, and multiplicities; locality makes Stokes applicable.
Crouzeix Endpoint	Amplified polynomial evaluation; rectangular compressions; Fourier coefficient energies.	Finite dual witnesses and positive blocks give a scalar cancellation; outer domains remove strict-containment assumptions.
Trace ideals Support result	Closed ideals; idempotent extended traces; positive cuts and scalar limits.	Algebraic recovery of ideals combines with canonical convergence to identify transported ideals.
All seams Support result	A shared splitting witness; named graded maps; rebased matrices and produced diagram values.	Eight local seam families cover the constructed diagram; equality transport retains its named maps.

3 Six mathematical interfaces

The cases are arranged by the mathematical work performed by the bridge. Each account identifies the objects on both sides, the preservation statement, and the point at which it is used. Source links refer to the inspected commit; the research files provide the larger declaration ledger.

3.1 Common bases: from a matroid representation to physical random tapes

The common-bases development connects an approximate counting argument to a fixed oracle machine, including its bit operations and the tape positions from which it obtains randomness [28]. Its organizing choice is to give the mathematical estimator, the finite random experiment, and the compiled program separate semantics, then prove the correspondences between them.

The mathematical problem is to count the common bases of two equal-rank matroids M_1, M_2 on an enumerated ground set $E = \text{Fin } n$. The first representation replaces two simultaneous base conditions by one base condition. On $E \sqcup E$, set

$$N = M_1 \oplus M_2^*, \quad T(A) = \text{inl}(A) \cup \text{inr}(E \setminus A).$$

The source proves that $T(A)$ is a base of N exactly when A is a base of both input matroids.¹ Equal ranks give $r(N) = n$, while every transversal $T(A)$ has size n . Consequently the deficiency

¹Base correspondence and the rank of the paired matroid.

$d(A) = n - r_N(T(A))$ vanishes precisely at the objects being counted. Explicit equivalences between subsets, finite subsets, and transversals transfer both cardinalities and partition sums. If Z is the common-base count, the resulting annealing partition function satisfies

$$Z \leq \sum_{A \subseteq E} q^{d(A)} \leq Z + 2^n q, \quad 0 \leq q \leq 1.$$

This representation identifies the quantity approximated by the sampling argument. It also supplies a useful state space for exchanges: a plain state selects one element from every doubled pair, while a defect state has one empty pair, one full pair, and singleton choices elsewhere. The walk can pass through these paired defects; a hitting construction returns to plain states. The formalization can therefore distinguish the space supporting local moves from the transversals whose weighted count is ultimately estimated.

The next correspondence replaces rank access by independence queries. A greedy oracle routine computes the rank of an enumerated subset; its correctness identifies the oracle-defined deficiency and annealing weights with their mathematical counterparts. The estimator also uses a sampled zero-deficiency witness as a final guard. This gives the exact zero-count behavior later required of every machine execution.

Finite random experiments are represented by trees whose internal nodes choose uniformly from a finite index set. A bounded implementation supplies each draw with a finite bank of rejection trials. For a tree with at most M successive draws and D trials per draw, the development bounds the loss in a $[0, 1]$ -valued observable by $M2^{-D}$. A separate relation permits permutations of the outcome indices at each node and proves that these reindexings preserve expectations. Thus the abstract exchange walk and its code can use different finite enumerations while retaining the same output law.

The code layer makes randomness explicit input. A typed expression language handles words, natural numbers, products, lists, folds, arithmetic, and oracle queries. An effect is an ordinary expression that takes an input and a random bank, and returns an optional value together with the unused bank. Its realization theorem relates execution on every bank to the corresponding random tree. For a Metropolis step, this theorem is assembled from the actual exchange proposal, acceptance calculation, Bernoulli experiment, and lazy branch. The complete estimator composes phase histories with the witness guard.

These expressions are then compiled into stack-machine blocks. The constructed program has nine registers and instructions for pushing, popping, taking a coin branch, querying an oracle, and halting. Its source evaluation semantics returns a value and a charge. A structural compiler-correctness proof shows that successful evaluation yields the same encoded output within that charge, including a separate simulation argument for list folds. A recursive certificate bounds intermediate encoding widths; expression-dependent polynomials then bound machine steps, oracle calls, and bit operations. The width certificate is combined with successful source evaluation: it controls the sizes visited by computation, while the realization and output certificates establish what is returned. Keeping these obligations separate makes the compiler theorem reusable for the deterministic preamble and the bank-processing postamble.²

The final probabilistic interface is especially revealing. Let V be the canonical bounded experiment, and let v be the rational output function implemented by the postamble. The certificate asserts

$$\mathbb{E}_w \varphi(v(w)) = \mathbb{E}_b \varphi(V(b)) \quad \text{for every } \varphi : \mathbb{Q} \rightarrow \mathbb{R},$$

²Cost-bearing compiler correctness; polynomial majorant.

where both expectations are finite uniform averages. It also records source execution and nonnegativity, including zero output when $Z = 0$.³ The universal observable interface transports the output distribution through changes of enumeration and bank layout.

The machine correspondence reaches actual tape addresses. For m total gathered bank bits, there is a start offset s , fixed before quantifying over tapes, such that every tape $\beta : \mathbb{N} \rightarrow \{0, 1\}$ produces

$$v(\text{physicalWord}(s, m, \beta)), \quad \text{physicalWord}(s, m, \beta)(i) = \beta(s + 3(m - 1 - i) + 1), \quad i < m.$$

The reversal comes from pushing bits onto a stack; the stride three reflects the sampling loop's instructions. The proof supplies $s + 3m + 1 \leq B$. Injectivity of these positions shows that a uniform length- B tape induces a uniform length- m word.⁴ Here m counts bank bits, whereas B bounds complete machine execution.

The endpoint chooses one program and constants C, a before the input matroids, oracle, and rational parameters $0 < \varepsilon, \delta < 1$. For every exact independence oracle and matroids of full ground set and common supplied rank r , it uses

$$B = C(1 + n + \ell + \lceil \varepsilon^{-1} \rceil + \text{clog}_2 \lceil \delta^{-1} \rceil)^a,$$

where ℓ is the encoded input length and clog_2 is the ceiling binary logarithm. Every infinite tape halts by B , with both resource counters at most B . For a length- B word, $\text{out}(w)$ denotes this output using its zero extension as the tape. Specializing the observable to the accuracy indicator gives

$$\#\{w \in \{0, 1\}^B : (1 - \varepsilon)Z \leq \text{out}(w) \leq (1 + \varepsilon)Z\} \geq (1 - \delta)2^B.$$

The specification packages this success count with the universal resource and zero-output guarantees.⁵ Verified compilation and probabilistic semantic preservation are established methods [5, 6, 23, 24]. Expectation-functional semantics for randomized programs is also established [1]. In Lean, SampCert separates sampling procedures from normalized distribution semantics and proves exact sampler laws [15]. The distinctive evidence here is the concrete integration: the proof carries the counting quantity through oracle code, width bounds, and physical tape coordinates to an exact finite success count.

3.2 Graph switches: a local table and a fiberwise error direction

The switch-chain development connects a four-vertex graph representation to a spectral estimate [33]. At each step the chain stays put with probability $1/2$; otherwise it chooses uniformly a four-vertex set and an ordered pair of distinct perfect matchings on it. It replaces the first matching by the second when the removed edges are present and the added edges absent, and stays put otherwise. Each of the $6\binom{n}{4}$ proposals thus has probability $1/(12\binom{n}{4})$. Its endpoint assumes $n \geq 4$ and a graphical degree sequence d with $d(v) \leq n - 1$. It bounds the mixing time at total-variation threshold $1/4$ by $2n^8$ and, when there is more than one realization, gives spectral gap at least

$$\frac{1}{24n^2\binom{n}{4}}.$$

The disjoint-pair branch of its proof is organized around a small exceptional direction in conditional averaging.

³The output certificate.

⁴Physical tape positions and their uniform law; the machine correspondence.

⁵The fixed-program and finite-tape specification.

States are simple graphs on $\text{Fin } n$ realizing d . Fix four distinct vertices, grouped into pairs $a = \{i, j\}$ and $b = \{k, l\}$. A coarse cell fixes the outer graph, the internal edge of each pair, and every outside vertex's total incidence into each pair. The remaining data consist of a 2×2 Boolean matrix M of cross edges and two sets U, W recording the singleton boundary choices at i and k . The source proves that these coordinates determine the graph.⁶

The important feasibility step reduces all four degree equations to preservation of the cross-edge total and two margin equations:

$$\text{row}_0(M) + |U| = c_L, \quad \text{col}_0(M) + |W| = c_R,$$

where c_L, c_R are fixed by the original graph. Explicit forward and inverse constructions identify the fixed-degree graph cell with left and right boundary data together with a compatible matrix. This equivalence preserves the multiplicities needed for uniform conditional averages. In particular, projecting a uniform graph cell onto its boundary data produces weights equal to the number of compatible cross matrices. The matrix count records the entire dependence between the left and right data that remains after the outer graph and incidence totals have been fixed.

There are only sixteen Boolean matrices. Let $c_s(r, t)$ count those with total s , first-row sum r , and first-column sum t . For $s = 2$, finite decision proves the table

$$(c_2(r, t))_{r, t=0}^2 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad c_2(r, t) = 2 - p(r) - p(t),$$

where $p(r) = \mathbf{1}_{r \neq 1}$.⁷ For every other total, the table factors into independent margin weights. For total two, the centered covariance factors through one flag on each side: conditional averaging can depend on the left data through the single Boolean flag $p(r)$. This is the dependence that the subsequent analysis tracks.

A short calculation illustrates the exceptional correlation. Among the six uniformly chosen total-two matrices, each off-center margin flag has mean $1/3$, while their product is always zero. Their covariance is therefore $-1/9$. This calculation concerns the uniform six-matrix experiment; graph cells also carry boundary multiplicities. It explains why a blanket claim of independent margins or nonnegative local correlation would fail and why the proof isolates a compensated error direction.

In the finite coordinate model, composing the two conditional averages therefore gives $A + Bp(r)$, with $B = 0$ unless $s = 2$. The graph-cell equivalence transports this formula back to actual graph states. This transport includes a proof that corresponding fibers have equal cardinalities and corresponding sums, and an identification of the flag: $p(r)$ says that i has equal adjacency bits to k and l , meaning both cross edges are present or both are absent.⁸

Write E_a for averaging over a pair-resampling fiber. The transported formula expresses the relevant conditional image as a fiber-constant base plus a fiber-constant weight times the equality flag; the weight is supported where the cross total is two. For this purpose a fiber is active when k and l each have exactly one neighbor in a . Subtracting E_a leaves a multiple of the centered flag on each active fiber. These functions form an error subspace whose orthogonal projection is denoted R_{ab} . The description is at most one dimensional on each fiber; the complete graph-space subspace can have many independent fiber components.

⁶The coarse graph cell; degree feasibility in coordinates.

⁷The finite total-two count.

⁸The affine conditional image; its graph interpretation.

This representation supplies the hypotheses of an abstract projection interaction inequality. Conditional averaging is a symmetric idempotent operator, and the error projection fixes the correlated fluctuation just identified. The analytic argument decomposes the state-space function using ranges and orthogonal complements of these projections. This turns the graph-specific conditional formula into precisely the range conditions required by the operator theorem. With G_{ab} the appropriate switch-gain projection, its graph instance yields

$$2\|G_{ab}f\|^2 - \|R_{ab}f\|^2 - \|R_{ba}f\|^2 \leq 2\langle (I - E_a)f, (I - E_b)f \rangle.$$

A separately proved projection budget bounds the sum of error norms by the sum of gain norms. Summing the compensated inequality over ordered disjoint two-element vertex sets a, b then gives

$$\sum_{a,b: a \cap b = \emptyset} \langle (I - E_a)f, (I - E_b)f \rangle \geq 0.$$

Thus the proof controls the aggregate interaction by accounting explicitly for the exceptional flag directions.⁹

The nonnegative disjoint sum combines with triple-generator and local comparison estimates to produce a generator-square bound, then the stated spectral gap and mixing estimate. Computation on finite objects and transport through equivalent representations are established proof methods [11, 21]. The case-specific mathematical choice is to isolate the total-two table's equality flag, identify its centered error space on graph fibers, and compensate those errors in a global operator estimate.

3.3 Mahler: ordered forms and boundary locality

The symmetric Mahler development connects exterior algebra, complex analysis, and convex geometry through several representations of the same differential form. Its selected endpoint, `OAI.SymmetricMahler.symmetric_mahler`, states

$$\frac{4^n}{n!} \leq |K| |K^\circ|$$

for $n \geq 1$ and a compact, convex, origin-symmetric set $K \subseteq \mathbb{R}^n$ with nonempty interior. Here volume is ordinary Lebesgue volume and the coordinate polar is literally $K^\circ = \{p : \sum_i p_i v_i \leq 1 \text{ for every } v \in K\}$. The accompanying mathematical argument uses holomorphic maps associated with finite intersections of strips [31]. For the architectural review, the instructive point is how a local transgression identity becomes a numerical flux value usable by the convex-body reduction.

The generic analytic results take a structure named `MassHypotheses`. Its fields require an open domain containing zero, holomorphic components f_j , zero as their unique common zero in the domain, a homogeneous degree- m leading polynomial map G without nonzero common zeros, an actual Euclidean Taylor remainder $f - G = O(\|z\|^{m+1})$, and compact closures of sublevels with $0 < R < 1$ contained in the domain. The structure records analytical inputs. The desired mass inequality and numerical flux value are separate conclusions. For the actual strip maps, `euclideanSpecial_massHypotheses` constructs every field from positive dimension, an injective real measurement matrix, and $m \geq 2$. The final body theorem supplies the resulting homogeneous flux value to a conditional reduction lemma. Its retained inputs are precisely the body hypotheses stated above.¹⁰

⁹The concrete interaction inequality; assembly with the projection budget.

¹⁰The endpoint and concrete hypothesis construction are `Mahler/MainTheorem.lean`, lines 12–22, and `Mahler/EuclideanSpecialMap.lean`, lines 67–97.

Write the complex dimension as $n = k + 2$. The deformation involves one-forms α_t and β , with an ordered primitive

$$\eta_t = \alpha_t \wedge \beta \wedge (d\alpha_t)^k.$$

The ambient differentiation theorem preserves both terms:

$$d\eta_t = \beta \wedge (d\alpha_t)^{k+1} - \alpha_t \wedge d\beta \wedge (d\alpha_t)^k.$$

It follows from actual Fréchet differentiation, exterior product rules, and $d^2\alpha_t = 0$. Restriction to sphere tangents then kills the first term. The deformation residual has $k + 1$ possible placements of $d\beta$; their verified multiplicity gives

$$\text{residual}_t = -(k + 1)d\eta_t \quad \text{on sphere tangent tuples.}$$

Keeping the ambient identity separate from its tangential consequence makes the vanished term, sign, and multiplicity individually accountable.

The disappearance of the first term has a concrete linear-algebra proof. At $z \neq 0$, the sphere tangent space has real dimension $2n - 1$ and contains the nonzero circle tangent iz . Homogeneity makes β and $d\alpha_t$ horizontal in this direction: inserting iz in any slot gives zero. Their top-degree wedge therefore vanishes. The lemma `horizontal_top_eq_zero` proves the general assertion explicitly. An independent top-degree tuple determines a basis. Choose a nonzero basis coordinate of the horizontal vector, replace the corresponding tuple entry by that vector, and expand it in the basis. Alternation kills every term except the selected coordinate times the original evaluation; cancelling the nonzero coordinate proves that evaluation zero. A dependent tuple already vanishes by alternation. The sphere theorem instantiates this argument with the verified tangent dimension and iz .

The representation seam occurs because the development constructs η_t in two useful ways. One bundles a literal shuffle form through finite basis expansion, supporting regularity proofs by evaluation. The other builds a continuous product from a two-form and continuous wedge powers, supporting differentiation rules. Recursive slot types retain each factor's ordered pair of arguments. Two equivalences flatten these slots to $\text{Fin}(2(k + 1))$ along different routes. A small extensional proof identifies the equivalences; injectivity of the projection from continuous alternating maps to alternating maps then identifies the entire fields. This field equality transports the proved transgression into the analytic theorem expecting the other representation. Thus the intermediate theorem `homogeneousFlux_endpoints_of_transgression` has an explicit local-formula premise, which `homogeneous_flux_endpoints` discharges.¹¹

Each representation carries different proof obligations. The literal shuffle description keeps permutation signs and factor order available to algebraic rewrites. The continuously bundled field has a normed-space structure, so its regularity and derivative can be expressed through the existing calculus API. Finite basis expansion connects these obligations: smoothness of the finitely many evaluated coefficients supplies smoothness of the bundled form. The source obtains every finite regularity order on punctured space from the analytic hypotheses, including the C^2 order used below. Identifying the fields also identifies their derivatives as functions valued in continuous alternating maps. This is stronger than checking their values on a chosen tangent tuple, which suffices only at the later integration boundary. The proof therefore uses two distinct congruence principles at appropriate stages: whole-field equality for differentiation and tangential agreement for oriented density. Keeping those principles separate clarifies which information each mathematical operation consumes.

¹¹See `Mahler/HomogeneousEndpoint.lean`, lines 12–45. The underlying tangent identity is proved in `HomogeneousTransgression.lean`, lines 65–97.

A second interface controls locality. The primitive is C^2 on punctured space, while the flux argument needs regularity near the sphere. `exists_closed_set_form_extension` takes a form ω that is C^2 on an open set U containing a closed set K . Smooth separation gives a function χ equal to one near K and zero near U^c . The field $\tilde{\omega} = \chi\omega$ is globally C^2 : use the product rule inside U and its eventual equality to zero outside U . Near K it is eventually equal to ω , so their Fréchet derivatives, and hence exterior derivatives, agree there. For K the sphere, coordinate Stokes applies to $d\tilde{\omega}$ and $d^2\tilde{\omega} = 0$ gives zero boundary flux. Tangent-density congruence then transfers that conclusion to the original residual.¹²

Zero residual flux makes the deformation's path flux constant by the fundamental theorem of calculus. Comparing endpoints supplies $(\pi m)^n$; the small-sphere limit and special strip-map reduction use that value in the body inequality. Dimension one is handled by a separate circle calculation. The mathematical devices are familiar. Mathlib already provides continuous alternating maps, exterior differentiation, and $d^2 = 0$ under smoothness assumptions [36]. Kudryashov's earlier Lean formalization already transfers a divergence theorem between integration theories and derives Cauchy–Goursat under carefully chosen differentiability assumptions [22]. Its rectangular domains give a direct comparison for integration interfaces. The distinctive step here identifies ordered exterior algebra with continuously bundled fields and combines that equality with a locality theorem that imposes regularity where the integral uses it. These interfaces expose and discharge the substantive premises without requiring regularity at the singular center.

3.4 Crouzeix: compression, energy, and strict neighborhoods

The finite-matrix Crouzeix endpoint states an amplified norm estimate with constant two. For an n -by- n complex matrix A , m -by- m coefficients B_k , and $n, m > 0$, `OAI.DirectCrouzeix.complete_e_crouzeix` states

$$\left\| \sum_{k=0}^d A^k \otimes B_k \right\| \leq 2 \sup_{z \in W(A)} \left\| \sum_{k=0}^d z^k B_k \right\|.$$

The source uses the operator 2-norm, literal matrix Kronecker products, and $W(A) = \{\langle u, Au \rangle : \|u\| = 1\}$. Its endpoint quantifies over arbitrary A and coefficient matrices. The accompanying argument [30] is organized around boundary analysis; the formal implementation makes particularly clear how positivity and Fourier energy combine to produce the constant two.

The local interface `analytic_domain_core` accepts a polynomial sequence b_k , a continuous boundary parametrization, and a matrix resolvent field $R(t)$. Its hypotheses give finite Faber expansions, lower and upper coefficient-energy bounds, exact coefficients $\hat{R}(-k) = b_k(A)$, vanishing positive Fourier coefficients, and $R + R^* \geq 0$. These are conditional inputs to the local theorem. `exterior_collar_bound` constructs them from exterior conformal data and supporting halfplanes before applying the core. In particular, the positive resolvent and coefficient bounds are outputs at this geometric stage, rather than retained assumptions of the public endpoint.

For the normalized core, assume $\|F(h(t))\| \leq 1$ on the boundary, where h is its parametrization. Let T be the amplified polynomial evaluation and $\gamma = \|T\| > 0$; the zero case is separate. A unit singular pair satisfies $Tx = \gamma y$ and $T^*y = \gamma x$. The supporting lemma obtains a norm attainer by compactness, then proves the adjoint relation through a squared-distance calculation. Reshaping vectors indexed by $\text{Fin}(n) \times \text{Fin}(m)$ produces rectangular matrices X, Y . All boundary integrals below use normalized circle measure. The proof forms

$$P = X^*RX, \quad Q = Y^*RY, \quad M = Y^*RX, \quad L = X^*RY, \quad a = \int \|M(t)\|_{\text{HS}}^2 dt.$$

¹²The extension and exact-form flux theorem are `Mahler/ClosedBoundaryStokes.lean`, lines 18–63.

A cross-boundary pairing estimate proves $a > 0$. The two singular relations produce the diagonal bounds

$$\int \|P + P^*\|_{\text{HS}}^2 \leq \frac{4a}{\gamma^2}, \quad \int \|Q + Q^*\|_{\text{HS}}^2 \leq \frac{4a}{\gamma^2}.$$

Fourier identities give $a \leq \int \|M + L^*\|_{\text{HS}}^2$. Compressing $R + R^*$ by the concatenated columns $[X \ Y]$ yields a positive block matrix with these diagonal and off-diagonal blocks. A trace estimate then gives

$$2 \int \|M + L^*\|_{\text{HS}}^2 \leq \int \|P + P^*\|_{\text{HS}}^2 + \int \|Q + Q^*\|_{\text{HS}}^2.$$

Thus $2a \leq 8a/\gamma^2$, and cancellation of $a > 0$ gives $\gamma \leq 2$.¹³

Two details explain why this reduction works sharply. First, `singular_diagonal_estimates` tests both ordered products FG and GF . The forward and adjoint singular relations turn them into different diagonal compressions, preserving the noncommutative order of matrix coefficients. The dual step is explicit: from bounds on every finite test family C_k , `weighted_coefficient_dual` substitutes $C_k = w_k^{-1} \overline{P_k}$ in the project's bilinear entry pairing. Both pairing and test energy become $s = \sum_k \|P_k\|_{\text{HS}}^2 / w_k$, leaving $s^2 \leq bs$ and hence $s \leq b$. Nonnegative bounded partial sums subsequently give the infinite sequence estimate. This keeps the optimizing witness finite and concrete.

Second, the weights are derived by boundary analysis. A Faber series splits into a nonnegative Fourier series and a remainder with no nonnegative coefficients. Orthogonality gives its energy as the sum of coefficient energy and remainder energy. The remainder is also the action of a nonnegative kernel with both marginals one on the nonconstant series. A weighted squared-norm inequality and Fubini prove kernel contraction. Consequently $w_0 = 1$ and $w_k = 2$ for $k > 0$. The resulting half-weighted coefficient tail fits the exact Fourier energy of $P + P^*$; the constant coefficient receives its own factor four.

A further interface supplies matrix-valued Faber expansions from scalar polynomials. A scalar sequence with its k th polynomial of degree k is mapped into polynomials over the matrix coefficient ring by sending scalars to scalar matrices. Its leading coefficients are units. The implementation then applies Mathlib's degree-filtered spanning theorem for polynomial sequences and extracts a finite matrix-coefficient witness from membership in that span. This gives the exact finite expansion consumed by the core. Evaluation and Kronecker-evaluation identities carry the witness to the two concrete matrix calculations. The established polynomial basis theorem is reused across the change of coefficient ring, while explicit transport lemmas preserve the order of multiplication. Together with rectangular reshaping, this explains how the scalar boundary analysis supplies estimates uniformly in the matrix coefficient size, the defining feature of the complete bound.

The final interface removes strict-domain geometry. For every compact, nonempty convex $K \subseteq \mathbb{C}$, `exists_exponential_outer` constructs a compact sublevel of a finite sum of exponentials of affine support inequalities. A finite direction net controls approximation; four coordinate directions ensure boundedness. The sublevel contains K strictly and lies in any prescribed neighborhood of K . In the endpoint, set $M = \sup_{W(A)} \|F\|$ and $\rho = M + \varepsilon/2$. Compactness places such an outer sublevel inside the open set $\{z : \|F(z)\| < \rho\}$. The scaled domain estimate gives $\|F(A)\| \leq 2M + \varepsilon$, and removing $\varepsilon > 0$ gives the desired bound. The final assumptions are positive matrix sizes and arbitrary coefficients; smooth strict containment is supplied by an auxiliary construction.¹⁴

Singular vectors, dual witnesses, positive blocks, and approximation are established tools. Their precise composition is the architectural lesson: amplification becomes rectangular compression, the

¹³The four compressions and closing cancellation appear in `DirectCrouzeix/DomainCore.lean`, lines 15–117.

¹⁴See `DirectCrouzeix/OuterApproximation.lean`, lines 59–122, and `CompleteBound.lean`, lines 65–97.

two polynomial orders become two energy bounds, and positivity closes a scalar cancellation. As a comparison, the Crouzeix–Palencia proof centers the Cauchy transform of conjugate holomorphic functions and obtains the complete constant $1 + \sqrt{2}$ [13]. The comparison identifies different proof interfaces, without asserting priority for the individual devices used here.

3.5 Recovering ideals from extended traces and canonical limits

The trace-cone development reconstructs actual closed ideals from an analytic invariant and then proves that an equivalence of invariants transports two distinguished ideals. The mathematical setting is the cone of extended, lower-semicontinuous traces, studied as an invariant of C^* -algebras by Elliott et al. [17]. The selected support results concern this ideal correspondence. Their hypotheses are nonunital complex C^* -algebras with compatible order and star structures; they impose no density condition on the finite domain of a trace [34].

An `ExtendedTraceA` is a function on positive elements with values in $[0, \infty]$, together with proofs of vanishing at zero, additivity, positive homogeneity, traciality, and lower semicontinuity. For a closed two-sided star ideal I , the code constructs the extended trace

$$\iota_I(a) = \begin{cases} 0, & a \in I, \\ \infty, & a \notin I, \end{cases} \quad a \geq 0. \quad (1)$$

This requires mathematical proofs, rather than only a convenient wrapper. For positive a, b , heredity of closed ideals gives $a + b \in I$ exactly when both summands belong to I , proving additivity. The equivalence $x^*x \in I \iff x \in I$ and star invariance give traciality; closed sublevel sets give lower semicontinuity. The existence theorem in `FiniteIdeal.lean` therefore supplies a trace satisfying the full interface, from which the later definition chooses a witness.

The reverse construction starts with the finite-square domain of a trace:

$$\mathcal{N}_\tau = \{x \in A : \tau(x^*x) < \infty\}, \quad I_\tau = \overline{\mathcal{N}_\tau}^{\|\cdot\|}. \quad (2)$$

The source proves that $\mathcal{N}_\tau$ is a nonunital star subalgebra absorbing multiplication on both sides, then takes its topological closure. Thus I_τ denotes a concrete closed ideal. If $\tau + \tau = \tau$, every value is either zero or infinity. Finite-trace positive elements approximate every positive element of I_τ ; lower semicontinuity extends their zero values to that limit, yielding $\tau = \iota_{I_\tau}$. Conversely, $I_{\iota_I} = I$. These inverse laws establish `closedIdealEquivIdempotent`, an equivalence between closed ideals and additive-idempotent traces.

The ideal order is also encoded by addition:

$$I \subseteq J \iff \iota_I + \iota_J = \iota_I. \quad (3)$$

Consequently, any additive bijection e of the two extended-trace cones already induces an order isomorphism Φ of their closed ideals, with $\iota_{\Phi(I)} = e(\iota_I)$. Neither continuity nor compatibility with scalars is required for this part. The subsequent analytic argument identifies which ideals this order isomorphism assigns to a given trace. This separation makes the strength of each interface field visible.

The relevant topology cannot be replaced silently by pointwise convergence. Write $(a - \varepsilon)_+$ for the positive cut defined by continuous functional calculus. The development proves

$$a \in I_\tau \iff \forall \varepsilon > 0, \quad \tau((a - \varepsilon)_+) < \infty, \quad a \geq 0. \quad (4)$$

Its canonical convergence predicate for a filtered family (τ_i) and a candidate limit σ requires, for every positive a and every $\varepsilon > 0$,

$$\limsup_i \tau_i((a - \varepsilon)_+) \leq \sigma(a) \leq \liminf_i \tau_i(a). \quad (5)$$

The two inequalities have different roles: the left regularizes the argument by a positive cut; the right uses the original element. Their combination, together with recovery of a lower-semicontinuous trace from its cuts, gives uniqueness of a canonical limit along a proper filter.¹⁵

In this sense the source proves

$$r\tau \longrightarrow \iota_{I_\tau} \quad (r \downarrow 0), \quad r\tau \longrightarrow \iota_{Z_\tau} \quad (r \rightarrow \infty), \quad (6)$$

where Z_τ is a closed ideal whose positive elements are exactly those on which τ vanishes. The first limit depends on the closure and cut characterization in (4). Whenever $a \in I_\tau$ but $\tau(a) = \infty$, every positive scalar multiple still has value $(r\tau)(a) = \infty$, although $\iota_{I_\tau}(a) = 0$. A claimed pointwise scalar-zero limit would therefore give the wrong assertion. Finite values on all positive cuts reconcile this behavior with (5). The scalar-infinity limit, by contrast, is first proved pointwise using the trace that is zero on τ 's zero set and infinity elsewhere.

Finally, a **TraceConeEquiv** packages an additive bijection, preservation of zero and positive scaling, and canonical convergence in both directions. For each scalar limit, the endpoint proof maps the family through e , rewrites using scalar compatibility, and invokes uniqueness of canonical limits. Injectivity of $I \mapsto \iota_I$ then yields

$$\Phi(I_\tau) = I_{e(\tau)}, \quad \Phi(Z_\tau) = Z_{e(\tau)}.$$

The displayed endpoint proofs use forward convergence preservation; the inverse field belongs to the larger equivalence interface. The illuminating architecture is the chain from constructed ideal weights, through an algebraic reconstruction, to analytic recognition by limits. It makes the ideal correspondence an actual transported object, rather than an unspecified provider assumption. This is a specific use of the abstraction boundaries familiar from large formalizations [7, 12].

3.6 Named graded maps across all seams of a produced diagram

The character-variety case proves that values produced by a marked surface construction satisfy every seam equation. The selected theorem, **producedMarkedValues_allSeams**, starts with an existing field-valued solution of the original diagram. Its hypotheses include properness of the diagram, a maximal-rank condition on the facet at the chosen seam, and positive genus of the parent facet. The conclusion refers to the specific diagram and values constructed from that solution [26]. The key issue is preserving named graded data while changing coordinates and subdividing the surface.

The seam equation is stronger than equality of flags. For a graded index set with grade function a and a frame e , let

$$F_k^e = e\{u : u_i = 0 \text{ whenever } a(i) \geq k\}, \quad q_k^e(v) = (e^{-1}v)|_{\{i:a(i)=k\}}.$$

Here q_k^e retains the named coordinates of grade k . The predicate **SameFramedFlagae** requires

$$F_k^e = F_k^f, \quad q_k^e(v) = q_k^f(v) \quad \text{for all } v \in F_{k+1}^e, \quad (7)$$

¹⁵The convergence predicate is defined in **TraceCone/Basic.lean**, lines 65–69; uniqueness is proved in **IdealTransport.lean**, lines 177–190.

for every k . Thus the comparison retains the maps representing the successive graded pieces, rather than only their dimensions or underlying subspaces.¹⁶ In the produced seam equations, the two frames are obtained from evaluations of the seam's left and right matrix words. Losing these named maps during a change of basis would weaken the theorem's meaning. The one-dimensional frame example in Section 4.4 illustrates this distinction without the surface construction.

The intermediate construction records precisely the data needed to prevent that loss. An **IdentifiedBand** contains a common frame for a simultaneous splitting, a rank shape, and row and column identifications of its graded fibers. Its prefix and named-coordinate laws relate those identifications to the two input frames. The source first proves that such a record exists using the named band grades and rank equalities. It then chooses a witness. The opaque declaration **cutGeneratorWitness** has this construction as its body; opacity hides reduction, rather than adding an unproved existence assumption. Classical choice here supplies coherent mathematical data, without claiming an effective procedure for computing the frame.

Crucially, the same chosen record supplies the shape, row ranks, column ranks, and marked frames of the produced diagram. Those fields also determine its punctures and values. Independently choosing compatible dimensions would not provide these shared identifications. This illustrates the difference between packaging a structure and carrying a sufficiently strong specification through it. Records and algebraic interfaces are standard techniques [3, 18]; the case-specific payload is the simultaneous preservation of the named graded maps in (7).

The global seam proof is assembled from eight local families: the chosen old seam, the other inherited seams, the first and internal original seams, the last and internal mirror seams, and the positive and negative branch seams. Exhaustiveness is proved using a surjective equivalence describing the grafted diagram, followed by sum, endpoint, and Boolean cases. Consequently, the universal quantifier over new seams follows from a structural coverage theorem. In the inherited family, coordinate equivalences identify the new frames with the old named seam frame; grade-preservation lemmas make these equivalences suitable for the full predicate. The original solution then supplies the seam equation. The local proof transports coordinate maps, not merely a statement that the old and new ranks agree.

There remains a second transport at the endpoint. The general **markedCut_allSeams** theorem proves the eight-family result for the assembled marked-cut data. The produced values use matrices rebased along the equality of the selected seam rank and its parent-facet rank. **rebaseUnit** performs this equality-induced change of matrix size. The endpoint proves that the chosen and rebased generator matrices agree, uses the mirror-rebasing theorem, and transports the resulting equality through the side-value constructor and then **valuesFromPorts**. Finally, **framedSeamEquations_of_values_eq** transfers the general seam predicate to the exact produced values. These equalities tie the compatibility proof to the constructed representation.

The boundary assumptions also distinguish two nearby statements. The all-seams endpoint has boundary indices b_0, i_0 , but requires no proof that this occurrence is the chosen seam. The later **producedMarkedSolution**, which assembles surface, seam, and vertex equations into a complete solution, additionally assumes

$$D.\text{boundarySide}\langle D.\text{bandCutParent}(q), b_0, i_0 \rangle = \langle q, \text{none} \rangle.$$

That extra hypothesis is used for the surface component. Keeping these endpoints distinct exposes the exact scope of the selected support theorem. The architectural lesson is the combination of a common witness, exhaustive local families, and equality transport that preserves the named mathematical content through the final construction.

¹⁶See the definition of **SameFramedFlag**.

4 Worked examples: why the stronger interface matters

The following elementary arguments isolate mechanisms used in the case studies. Their purpose is to make the preservation requirements testable and understandable outside the full developments.

4.1 Injective tape positions preserve a uniform experiment

Proposition 1. *Let $p : \{0, \dots, m-1\} \rightarrow \{0, \dots, B-1\}$ be injective. If β is uniform on $\{0, 1\}^B$, then $(\beta(p(0)), \dots, \beta(p(m-1)))$ is uniform on $\{0, 1\}^m$. Every word has exactly 2^{B-m} preimages.*

Proof. Prescribing the m selected bits leaves precisely $B - m$ coordinates free. Each assignment to these remaining coordinates determines one preimage. Thus every word has the same number of preimages, and its probability is $2^{B-m}/2^B = 2^{-m}$. $\square$

For the common-base machine, the relevant map is $p(i) = s + 3(m - 1 - i) + 1$. Reversal and multiplication by three preserve distinctness. For $i < m$, its value is less than B whenever $s + 3m + 1 \leq B$. The latter bound is stronger than needed for the selected positions because it also accounts for the surrounding execution. The review’s Lean file proves these arithmetic facts for arbitrary natural s, m, B , including the empty-word case. Its index definition is

```
def physicalIndex (start m : Nat) (i : Fin m) : Nat :=
  start + 3 * (m - 1 - i.val) + 1
```

The proof uses the bounds stored in `Finm` to justify subtraction arithmetic. This matters: natural-number subtraction is truncated, so injectivity would fail on unrestricted natural indices. A range type is part of the mathematical interface, not a cosmetic annotation.

For comparison, if $p(0) = p(1)$, the words 01 and 10 have no preimages. Reading fair individual bits does not by itself produce an independent word. Proposition 1 explains why the source proves injectivity of the physical positions before transporting the success probability. The Python companion exhaustively checks 15 small choices of (s, m, B) against the exact fiber count.

4.2 Observables determine a finite law; its mean does not

Proposition 2. *Let μ, ν be probability measures on a finite set X . Then $\mathbb{E}_\mu \varphi = \mathbb{E}_\nu \varphi$ for every $\varphi : X \rightarrow \mathbb{R}$ if and only if $\mu = \nu$.*

Proof. For each $x \in X$, take $\varphi = \mathbf{1}_{\{x\}}$ to obtain $\mu(\{x\}) = \nu(\{x\})$. Summing singleton masses gives equality on every subset. The converse follows from the definition of finite expectation. $\square$

In a uniform finite experiment, these expectations are normalized sums with integer multiplicities. The Lean companion proves the corresponding count statement: equality of weighted sums against every natural-valued observable determines each multiplicity. It then checks the measures on $X = \{0, 1, 2\}$ with multiplicities

$$(1, 1, 1) \quad \text{and} \quad (0, 3, 0).$$

Both have total mass three and mean one after normalization. Their probabilities of $\{0\}$ are $1/3$ and zero. Thus preservation of the expected numerical output would be insufficient for an accuracy event. The universal observable certificate in Section 3.1 supports exactly the stronger conclusion required: substitute the indicator of the desired relative-error interval.

This interpretation also explains why the source allows reindexing of draw outcomes. A permutation can change the result returned for a particular numbered bank without changing

its distribution under uniform sampling. Law equality is the appropriate interface between those experiments; the deterministic compiler layer separately preserves the execution result for a supplied bank.

4.3 A cut-sensitive limit that is not pointwise

Consider the commutative C^* -algebra c_0 of complex sequences tending to zero, with pointwise operations and the supremum norm. On its positive cone define

$$\tau(a) = \sum_{j=1}^{\infty} a_j \in [0, \infty].$$

The trace is additive, positively homogeneous, and tracial. It is lower semicontinuous because it is the supremum of its continuous finite partial sums. Finite-support sequences belong to $\mathcal{N}_\tau$, so the closed finite ideal I_τ in (2) is all of c_0 .

For every positive $a \in c_0$ and $\varepsilon > 0$, the sequence $(a - \varepsilon)_+$ has finite support. Consequently

$$\lim_{r \downarrow 0} r\tau((a - \varepsilon)_+) = 0, \quad 0 \leq \liminf_{r \downarrow 0} r\tau(a).$$

These are precisely the two inequalities in (5) with limit trace $\iota_{c_0} = 0$. But for $a_j = 1/j$ one has $\tau(a) = \infty$, hence $r\tau(a) = \infty$ for every $r > 0$. The pointwise limit at this element is infinity. The finite ideal and canonical limit are nevertheless unambiguous.

This classical example makes the reason for the positive cuts concrete [17]. Closure admits elements with infinite trace, while every fixed positive cut is controlled by a finite value. Replacing the source's convergence predicate by pointwise convergence would therefore alter the theorem.

4.4 Local equality and named maps carry extra information

Two further short examples clarify the other cases. The real functions $f(x) = 0$ and $g(x) = x$ agree at zero but have different derivatives there. Equality on a neighborhood of zero does imply equality of their derivatives at zero. The Mahler cutoff argument uses eventual equality near the boundary to transfer the ambient derivative, whereas agreement on tangent arguments suffices for the later comparison of integration densities. Mathlib's differential-form interface explicitly uses the derivative of a form-valued field [36].

Likewise, two frames of a one-dimensional vector space can induce the same flag but different named coordinate maps. If $e(u) = uv$ and $f(u) = 2uv$ over a field of characteristic zero, both give the flag $0 \subset \text{span}(v)$, while $e^{-1}(v) = 1$ and $f^{-1}(v) = 1/2$. The all-seams predicate must therefore retain the named graded maps when those maps occur in the seam equation. These examples explain the extra information consumed by the source's ambient-derivative congruence and named-coordinate equations, even when the coarser objects agree.

5 What is distinctive, and what can be reused?

The six cases suggest four kinds of preservation requirement. Exact data must survive when a later construction uses named coordinates or a particular polynomial expansion. Probability laws must survive when the next statement concerns an event rather than an average output. Estimates must survive when error terms are summed or a constant is optimized. Local differential information must survive when a derivative or boundary integral is formed. A single development can require several of

these: the common-base proof uses both deterministic execution equality and distributional equality, while Mahler uses whole-field equality and then boundary-local agreement. An event probability factors through the output law, so reindexing can be harmless in the first case. A named-coordinate equation consumes the coordinate map itself; ambient differentiation consumes the field near the point. The operation to be performed determines which correspondence is sufficient.

This classification is useful because it identifies the failure caused by an interface that is too weak. A mean does not determine an accuracy probability. A finite table describes the graph averages only after its multiplicities have been related to those of the graph fibers; here an explicit bijection supplies that relation. Equal subspaces do not determine named quotient coordinates. Equality at boundary points does not establish the ambient derivative congruence used by this Stokes argument. Pointwise scalar-zero limits need not equal the weights of the closed finite ideals of extended traces. These are concrete mathematical failures, illustrated in Section 4, rather than generic objections to abstraction.

Several reusable patterns emerge. First, use a representation in which the local calculation is small, then prove an explicit correspondence to the original state space. The switch-chain matrix table is useful because the graph-cell equivalence identifies its counting measure and its conditional fibers. Second, give the semantic relation the strength of the downstream claim. The all-observables certificate handles changes of enumeration without imposing an unnecessary same-bank equality. Third, keep hypothesis discharge visible. The Mahler strip map constructs the analytic inputs; the Crouzeix exterior domain constructs the resolvent and coefficient estimates. Fourth, preserve a common witness across dependent constructions. The marked-cut diagram and its values use the same splitting data, so the final seam theorem refers to the actual produced object.

These patterns have substantial antecedents. Verified compilers prove semantic preservation and make the relationship between a source language and its implementation explicit [23, 24]. Probabilistic program logics already reason about output distributions and observations [5, 6]. Algebraic hierarchy design and specification-driven collaboration already treat interfaces as mathematical objects [3, 12, 18]. Accordingly, the evidence supports a claim of informative, concrete compositions of established methods. It does not establish a historical first use of transport, finite decision, dependent records, or compiler correctness.

What makes the selected steps worth studying is their mathematical specificity. The random tape map records a real execution convention and feeds an exact success count. The exceptional graph direction is identified by the total-two matrix table and controlled by a projection budget. The ordered form identity retains the sign and multiplicity needed for transgression. Rectangular compressions organize amplification so that two energy bounds close at the constant two. Positive cuts recover the ideal hidden by infinite trace values. Named graded maps keep the produced diagram's equations intact under rebasing. These choices explain more of the proofs than a description of tactics or file size would.

The review also has clear limits. Selection favors readable, explicit bridges, so it cannot estimate how frequent these techniques are across the collection or compare development effort with other projects. The source analysis establishes what the inspected declarations state and how their displayed constructions connect. A full assessment of the headline mathematical results additionally involves rebuilding their dependencies and examining the underlying mathematical arguments. Those tasks should retain the same endpoint distinctions used here, particularly for the two support results.

For a mathematician, the practical benefit is a route into the source: identify the next operation, ask what it consumes, and inspect the lemma that preserves that information. For a formalizer, the cases suggest interfaces worth extracting into smaller libraries or examples. Their value lies in making a mathematical dependency explicit enough to inspect, reuse, and test. The worked

examples demonstrate a modest version of that process without requiring the entire surrounding development.

Data and code availability

The accompanying source package contains the LaTeX manuscript, verified reference records, declaration evidence ledgers, original Lean examples, and Python checks. The Lean examples use Lean 4.34.1 and import only `Lean`; their compiler log records the axioms reported for the four named theorems. The Python checks enumerate tape fibers and Boolean matrix counts using exact integers. The package README gives reproduction commands and identifies the pinned external sources. Public source links in the manuscript resolve to the reviewed repository revision.

Generative AI disclosure

OpenAI Codex agents contributed substantially to source discovery, literature synthesis, and mathematical exposition.

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